

Multiscale Hydrodynamic Entropy in FourCastNet

A Spectral Analysis of AI-Learned Fluid Dynamics

Aaditya Raj Yadav
Department of Physics, IIT Kanpur

April 25, 2026

Domain: Deep Learning, Fluid Dynamics, and High-Performance Computing (HPC)

1 Introduction and Project Objective

Modern weather forecasting is undergoing a paradigm shift. Traditional Numerical Weather Prediction (NWP) integrates the explicitly known Navier-Stokes equations over time. In contrast, Machine Learning Weather Prediction (MLWP) models, such as FourCastNet, learn fluid dynamics implicitly by optimizing billions of parameters over massive archives of historical data (e.g., ERA5 reanalysis).

This project rigorously investigates whether the Adaptive Fourier Neural Operator (AFNO) architecture internally replicates the physical multiscale turbulence cascades inherent to Earth's atmosphere, or if it resorts to non-physical "latent" data compression strategies.

By intercepting the complex spectral activations during a full 24-hour global inference pass on real ERA5 data, and applying Professor M.K. Verma's formulation of **Hydrodynamic Entropy** (S_H), we aim to:

- Map the thermodynamic trajectory of the neural network from initial chaotic input to stable forecast (Inter-block analysis).
- Quantify the intra-block power-law scaling of the AI's internal energy cascade.

2 Tensor Mechanics & The Spectral Domain

FourCastNet separates spatial scales mathematically using the Fourier domain, operating on a high-resolution global grid.

2.1 Dimensions and Patch Embedding

The raw input weather state is a tensor $X \in \mathbb{R}^{B \times C \times H \times W}$:

- **B (Batch):** 4 (representing a 24-hour period evaluated at 6-hour intervals).
- **C (Channels):** 20 physical variables (e.g., U/V winds at 10m, Temperature, Geopotential).
- **$H \times W$:** 720×1440 latitude-longitude grid (0.25° spatial resolution).

The architecture first tokenizes the physical map into patches, projecting the 20 physical variables into a high-dimensional latent space (e.g., $C_{\text{hidden}} = 1024$ abstract thermodynamic variables).

2.2 The Forward Pass into Fourier Space

A 2D Real-to-Complex Fast Fourier Transform (rFFT) maps the spatial tokens to the spectral domain. Mathematically, for a given spatial grid $X(x, y)$, the discrete transform $\mathcal{F}$ yields the spectral state $Z(k_x, k_y)$:

$$Z(k_x, k_y) = \mathcal{F}(X)(k_x, k_y) = \sum_{x=0}^{H'-1} \sum_{y=0}^{W'-1} X_{\text{token}}(x, y) e^{-i2\pi \left(\frac{k_x x}{H'} + \frac{k_y y}{W'} \right)} \quad (1)$$

Because the tokens are strictly real numbers, the transform utilizes **Hermitian symmetry** ($Z(-k_x, -k_y) = Z^*(k_x, k_y)$), halving the longitude dimensionality to save memory.

2.3 Physical Length Scales

In the Fourier domain, the spatial frequencies (wavenumbers k_x, k_y) correspond inversely to physical distances on Earth. With the planetary circumference at the equator $L_{\text{eq}} \approx 40,000$ km:

- $k = 1$ (**Planetary Waves**): Represents massive, global-scale structures like Rossby waves ($\sim 40,000$ km).
- k_{max} (**Grid Nyquist Limit**): Represents localized, sub-grid turbulence down to the 0.25° resolution limit (~ 28 km at the equator).

2.4 Weights (W) vs. Activations (a)

Within the spectral domain, the AFNO applies a learned channel-mixing complex weight matrix W .

- **The Weights (W)**: W is uniformly broadcast (shared) across all spatial frequencies to prevent memory explosion. Evaluating the static weights yields a non-physical flat spectrum ($E_{\text{mode}}(k) = |W|^2 = \text{Const}$).
- **The Activations (a)**: To observe fluid dynamics, we strictly analyze the **Activations** $a(\mathbf{k}) = W \cdot Z(\mathbf{k})$. This dynamic tensor represents the actual state vector of the simulated atmosphere flowing through the network.

3 Mathematical Framework: Hydrodynamic Entropy

Based on Verma et al. (2024, PRE), Hydrodynamic Entropy (S_H) quantifies macroscopic, multiscale disorder. Unlike microscopic Boltzmann entropy, S_H is nonextensive and captures the multiscale nature of turbulent cascades.

3.1 Scalar Wavenumber Mapping & 1D Shell Energy

For every point in the 2D complex activation grid, we calculate its radial scalar wavenumber:

$$k = \sqrt{k_x^2 + k_y^2} \quad (2)$$

We compute the 1D Shell Energy $E(k)$ by summing the squared magnitudes of the complex activations over all hidden channels (c) and across all modes residing on the scalar radial ring k :

$$E(k) = \sum_{c=1}^{C_{\text{hidden}}} \sum_{|\mathbf{k}|=k} |a_c(\mathbf{k})|^2 \quad (3)$$

3.2 Verma’s Entropy Calculation

The Shell Energy is normalized to derive a probability distribution p_k :

$$p_k = \frac{E(k)}{\sum_k E(k)} \quad \text{where} \quad \sum p_k = 1 \quad (4)$$

The Hydrodynamic Entropy for an individual block’s activation state is defined as:

$$S_H = - \sum_k p_k \log_2 p_k \quad (5)$$

4 Computational Implementation & GPU Vectorization

To extract the physical fluid properties dynamically without crashing memory limits, the implementation executes a strict sequence of PyTorch operations on the real FourCastNet weights and ERA5 .h5 files.

4.1 “Wiretapping” Activations

The model is executed under `torch.no_grad()` to disable backpropagation. We inject PyTorch `register_forward_hook` functions into the architecture. As the ERA5 data flows through the model during the 24-hour prediction cycle, the hooks passively copy the complex output tensors of the spectral mixing layer in all 12 blocks.

4.2 GPU Vectorization: The 2D to 1D Collapse

Computing $E(k)$ requires mapping the 2D spatial frequencies to 1D radial bins. Using nested Python `for`-loops over the grid across the batch is computationally prohibitive. This is solved via massively parallel GPU tensor operations:

1. **Define Bins:** We discretize the continuous k field using $\Delta k = 1$. `torch.bucketize` assigns a 1D bin index to every pixel in the 2D grid.
2. **Vectorized Collapse (`scatter_add_`):** We initialize a zero-tensor. Using `torch.scatter_add_`, the GPU simultaneously routes and sums the modal energy ($|a(\mathbf{k})|^2$) of all pixels into their corresponding 1D radial bin indices, yielding the $E(k)$ Shell Energy tensor instantly.

5 Theoretical Expectations on Real Data

By analyzing the real pretrained weights against real ERA5 atmospheric inputs, we anticipate the following structural behaviors:

5.1 Intra-Block Spectrum: “Latent Whitening”

Physical atmospheric enstrophy dictates a steep turbulent decay (e.g., $E(k) \propto k^{-3}$). However, deep learning networks suffer from vanishing gradients if high-frequency variance approaches zero.

- **Expectation:** We hypothesize that the internal activations will deviate from physical reality. The AFNO will perform **Latent Whitening**—artificially boosting the energy of high frequencies (small-scale storms) to maximize the channel capacity of the hidden variables.
- **Plot Implication:** The log-log plot of $E(k)$ vs k will exhibit a much shallower (or even positive) slope compared to the physical k^{-3} reference line.

5.2 Inter-Block Trajectory: Chaos to Order

As the data propagates through the sequential blocks, the network repeatedly applies a Soft-Thresholding operator (S_λ) which acts as “informational viscosity.”

- **Expectation:** We hypothesize that S_H will start relatively high at Block 0 (reflecting the chaotic, multiscale variance of the raw ERA5 input) and **drop monotonically** as the depth increases.
- **Plot Implication:** A strict downward trajectory in S_H across the 12 blocks. This will empirically prove that FourCastNet operates as a dissipative fluid simulator, filtering out high-entropy sub-grid noise to organize the atmospheric energy into stable, predictable macroscopic structures.

6 Future Validation

The scripts implemented in this framework (`run_real_era5_entropy.py`) will generate diagnostic visualizations. Future work will involve directly comparing the expected deviations established in this report against the empirically measured slopes and entropy deltas computed by the GPU vectorization engine.